

Lecture Content – Unit 1: Introduction to Operating System

Topic 1: Types of Operating System

Course: Operating System (DI03016061)

Programme: Diploma Engineering – Information Technology

Duration: 60 Minutes

1. Hook / Introduction (≈ 5 Minutes)

Classroom Hook

Ask students:

"Suppose you are using WhatsApp on your smartphone, listening to music, downloading a file, and receiving a phone call—all at the same time. How does your phone manage all these activities smoothly?"

Allow students to think and answer.

Now ask another question:

"Can the operating system used in a hospital life-support machine be the same as the operating system used in a mobile phone?"

Students will realize that different situations require different operating systems.

Motivation

Just as different vehicles are designed for different purposes—a bicycle for short distances, a truck for transporting goods, and an airplane for long-distance travel—different **Operating Systems (OS)** are designed to solve different computing problems.

Today's lecture introduces the **Types of Operating Systems**, helping students understand where and why each type is used.

Learning Outcomes

At the end of this lecture, students will be able to:

- Define different types of Operating Systems.
- Differentiate among various operating systems.
- Identify suitable applications of each OS.
- Compare their advantages and limitations.
- Relate operating systems to real-world engineering applications.

2. Core Concepts ($\approx$ 40 Minutes)

What is an Operating System?

An **Operating System (OS)** is system software that acts as an interface between the user and computer hardware. It manages resources such as CPU, memory, storage devices, files, and input/output devices.

Different computing environments require different operating systems. Based on their working style and purpose, operating systems are classified into several types.

1. Batch Operating System

A **Batch Operating System** executes a group (batch) of similar jobs one after another without direct user interaction.

Working

- Users submit jobs.
- Jobs are collected into batches.
- The operating system processes them sequentially.

Example

- Salary processing
- Electricity bill generation
- University result processing

Advantages

- High throughput
- Suitable for repetitive tasks
- Reduces idle CPU time

Disadvantages

- No interaction during execution
- Long waiting time
- Difficult to debug

Analogy

Imagine depositing examination answer sheets. Teachers evaluate them later in one batch instead of checking them immediately.

2. Time-Sharing Operating System

A **Time-Sharing Operating System** allows multiple users to use the same computer simultaneously by allocating a small time slice to each user.

Examples

- UNIX
- Linux Server
- Cloud-based laboratory systems

Advantages

- Fast response
- Resource sharing
- Supports multiple users

Disadvantages

- Security challenges
- High system complexity

Analogy

Like a teacher giving each student one minute to ask questions during a classroom discussion.

3. Multiprogramming Operating System

In a **Multiprogramming OS**, several programs remain in memory simultaneously. When one program waits for input/output, the CPU executes another program.

Advantages

- Better CPU utilization
- Higher efficiency
- Reduced idle time

Example

Running MS Word, Chrome, and VLC Player together.

4. Multiprocessing Operating System

A **Multiprocessing OS** uses two or more processors to execute tasks simultaneously.

Examples

- Modern Intel Core processors
- AMD Ryzen systems
- Data center servers

Advantages

- Faster execution
- Improved reliability
- Better performance

Real-Life Example

A restaurant with four chefs preparing different dishes simultaneously.

5. Multitasking Operating System

A **Multitasking Operating System** enables a single user to perform multiple tasks at the same time.

Examples

- Windows 11
- macOS
- Android
- Ubuntu

Daily Life Example

Listening to music while preparing notes and downloading files.

Advantages

- Better productivity
 - Improved user experience
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6. Real-Time Operating System (RTOS)

A **Real-Time Operating System** guarantees that critical tasks are completed within a specified time.

Examples

- Air Traffic Control
- Medical Equipment
- Industrial Robots
- Missile Guidance Systems
- Anti-lock Braking System (ABS)

Types

- Hard Real-Time OS
- Soft Real-Time OS

Advantages

- High reliability
 - Predictable response
 - Used in safety-critical systems
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7. Network Operating System (NOS)

A **Network Operating System** manages computers connected through a network.

Examples

- Windows Server
- Linux Server
- Novell NetWare

Functions

- User authentication
- Printer sharing
- File sharing
- Network security

Real-Life Example

College computer laboratories where students access shared files and printers through a central server.

Comparison of Operating Systems

Type	Users	Main Feature	Example
Batch	Many jobs	Sequential execution	Payroll Processing
Time Sharing	Multiple users	Time slicing	UNIX
Multiprogramming	Multiple programs	Better CPU utilization	Linux
Multiprocessing	Multiple CPUs	Parallel execution	Windows Server
Multitasking	Single user	Multiple applications	Windows 11
Real-Time	Critical systems	Guaranteed response	Medical Equipment
Network	Network users	Resource sharing	Windows Server

Fun Fact

NASA spacecraft, modern fighter aircraft, self-driving cars, and even Mars rovers rely on specialized **Real-Time Operating Systems** because even a delay of a few milliseconds can lead to mission failure.

Suggested Classroom Visuals

Diagram 1: Classification of Operating Systems

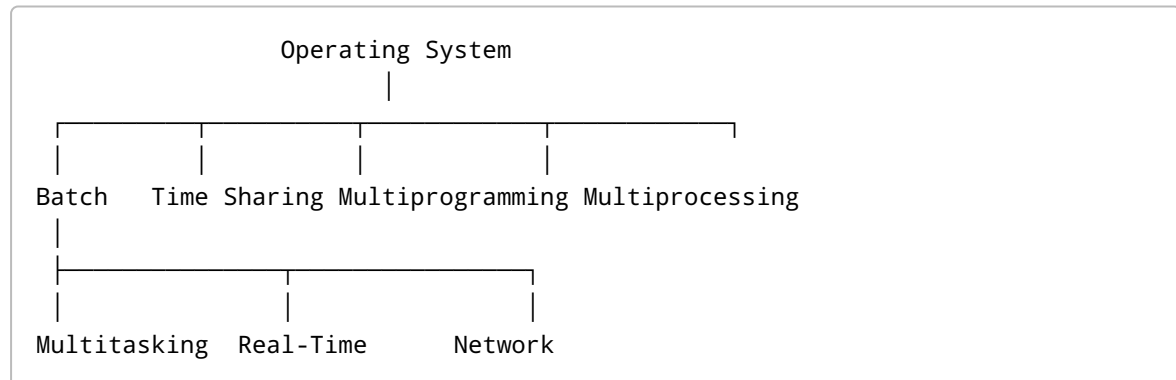


Diagram 2: Time Sharing Concept

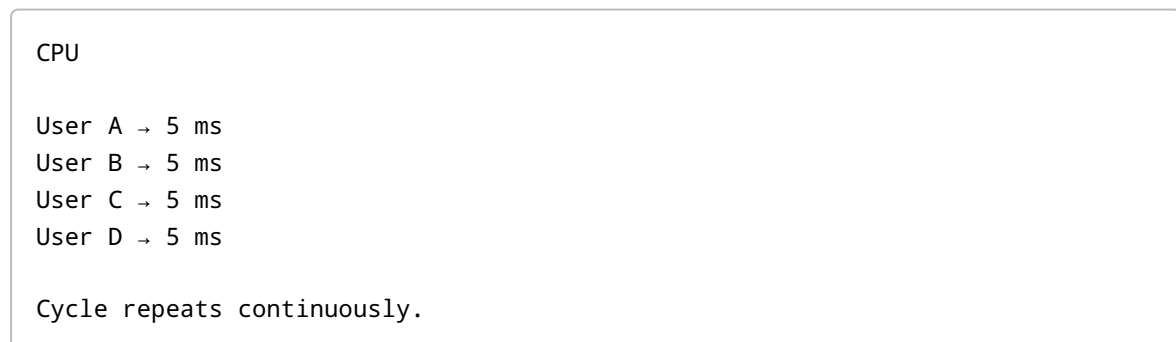
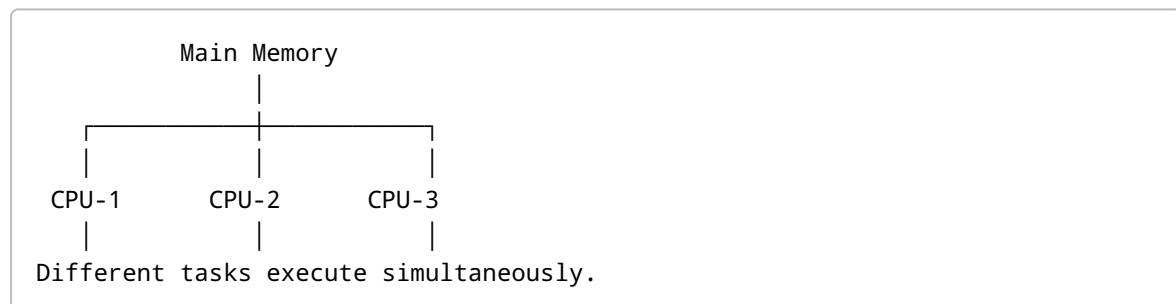
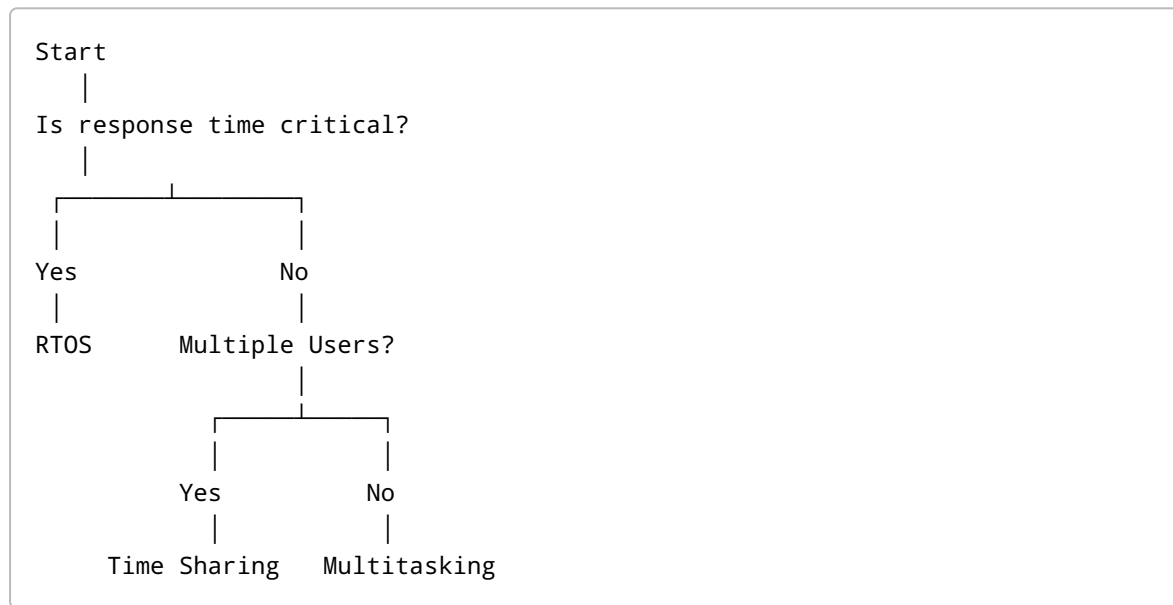


Diagram 3: Multiprocessing



Flowchart: Selecting an Operating System



3. Real-World / Industry Applications (≈ 10 Minutes)

Different operating systems serve different industries:

- **Batch OS:** Banking, payroll, utility bill generation, university examination processing.
- **Time-Sharing OS:** Cloud computing, university laboratories, software development environments.
- **Multiprogramming & Multitasking:** Personal computers, office automation, web browsing, multimedia editing.
- **Multiprocessing OS:** AI training, scientific computing, gaming, engineering simulations.
- **Real-Time OS:** Medical devices, autonomous vehicles, robotics, aviation, industrial automation.
- **Network OS:** Corporate offices, educational institutions, data centers, cloud servers.

Industry Insight:

Companies like Google, Amazon, Microsoft, and Meta use Linux-based servers with multiprocessing and network operating systems to manage millions of users worldwide.

4. Summary & Q&A (≈ 5 Minutes)

Key Takeaways

- Different computing needs require different operating systems.
- Batch OS processes jobs in groups.
- Time-Sharing OS serves multiple users efficiently.
- Multiprogramming improves CPU utilization.

- Multiprocessing increases performance using multiple CPUs.
- Multitasking enables users to run multiple applications simultaneously.
- Real-Time OS is essential for safety-critical applications.
- Network OS manages shared resources across computers.

Quick Revision Questions

1. What is a Batch Operating System?
2. Differentiate between Time-Sharing and Multitasking OS.
3. Why is RTOS used in medical equipment?
4. Give two applications of Network Operating Systems.
5. Explain Multiprocessing with an example.

Typical Student Doubts

- Is multitasking the same as multiprogramming?
- Can Windows support multiprocessing?
- Why is Linux widely used in servers?
- Which OS is used in mobile phones?
- Where is RTOS preferred over general-purpose operating systems?

Mentorship Note

Understanding the **Types of Operating Systems** is a foundation for advanced subjects such as **Computer Networks, Cloud Computing, Cyber Security, Embedded Systems, Internet of Things (IoT), Mobile Application Development, and Artificial Intelligence**. During internships and technical interviews, recruiters often ask scenario-based questions such as *"Which operating system would you choose for a hospital monitoring system?"* or *"Why are Linux servers preferred in cloud computing?"* Mastering this topic will strengthen your problem-solving ability, help you select appropriate technologies for engineering projects, and prepare you for careers as a **System Administrator, Software Engineer, Cloud Engineer, Network Engineer, DevOps Engineer, Embedded Systems Developer, or Cybersecurity Professional**.